

Laboratory Work No. 11

Emacs Text Editor

Mikhail Vzvarov — NPIbd-01-25

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Emacs Text Editor

Discipline: Operating Systems

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Group: NPIbd-01-25

Objective

- ▶ Become familiar with the Linux operating system.
- ▶ Obtain practical skills in working with the Emacs text editor.
- ▶ Learn basic Emacs keyboard commands.
- ▶ Practice working with buffers, windows, search and text editing.

Emacs Basics

Emacs is a powerful and extensible text editor.

Main concepts:

- ▶ **Buffer** — contains text or other information.
- ▶ **Frame** — the main Emacs graphical frame.
- ▶ **Window** — an area displaying a buffer.
- ▶ **Minibuffer** — used to enter commands and additional information.
- ▶ **Point** — the current cursor position.
- ▶ **Mode** — defines editing behaviour for a particular type of content.

Important Commands

C-x C-f	Open a file
C-x C-s	Save a file
C-x C-c	Exit Emacs
C-a	Beginning of line
C-e	End of line
M-<	Beginning of buffer
M->	End of buffer
C-k	Kill text to end of line
C-y	Yank text
C-x u	Undo

Creating and Opening the File

The practical work was performed in:

```
~/reports/lab11
```

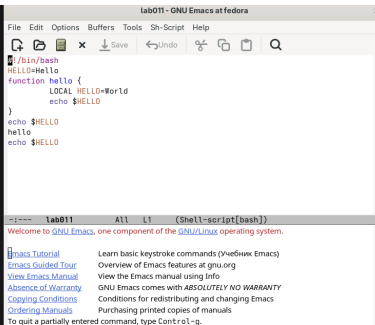
The created file was:

```
lab011
```

It was opened in Emacs with:

```
emacs lab011
```

```
1 #!/bin/bash
2 HELLO=Hello
3 function hello {
4     LOCAL_HELLO=World
5     echo $HELLO
6 }
7 echo $HELLO
8 hello
9 echo $HELLOmichailvzarov@fedora:~/reports/lab11$
michailvzarov@fedora:~/reports/lab11$ emacs lab011
```



Working with Buffers

Emacs can keep many buffers active at the same time.

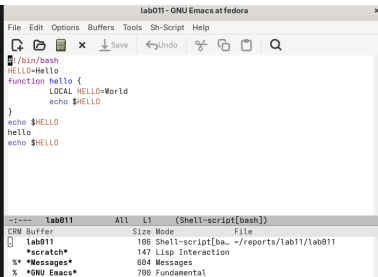
The buffer list was opened with:

C-x C-b

A different buffer can be selected with:

C-x b

```
1 #!/bin/bash
2 HELLO=Hello
3 function hello {
4     LOCAL HELLO=World
5     echo $HELLO
6 }
7 echo $HELLO
8 hello
9 echo $HELLO
michailvzarov@fedora:~/reports/lab11$ emacs lab011
```



The screenshot shows the GNU Emacs buffer list window titled "lab011 - GNU Emacs at fedora". The window has a menu bar with "File", "Edit", "Options", "Buffers", "Tools", "Sh-Script", and "Help". Below the menu bar is a toolbar with icons for file operations and editing. The main area displays a list of buffers. The "lab011" buffer is selected and highlighted. Below the buffer list, the content of the selected buffer is shown, displaying the same shell script as in the previous block.

Buffer	Size	Mode	File
lab011	106		Shell-script[ba... ~/reports/lab11/lab011
scratch	147		Lisp Interaction
%* Messages*	884		Messages
% GNU Emacs*	788		Fundamental

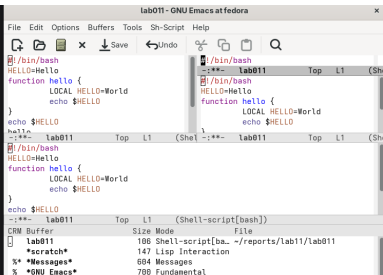
Working with Windows

The Emacs frame was divided into several windows.

Commands used:

- C-x 3 Split vertically
- C-x 2 Split horizontally
- C-x o Select another window
- C-x 1 Keep only the current window

```
1 #!/bin/bash
2 HELLO=Hello
3 function hello {
4     LOCAL HELLO=World
5     echo $HELLO
6 }
7 echo $HELLO
8 hello
9 echo $HELLO
michailvzarov@fedora:~/reports/lab11$ emacs lab011
```



Major mode
mouse-1: Display major mode menu
mouse-2: Show help for major mode
mouse-3: Toggle minor modes

Text Search

Forward incremental search was started with:

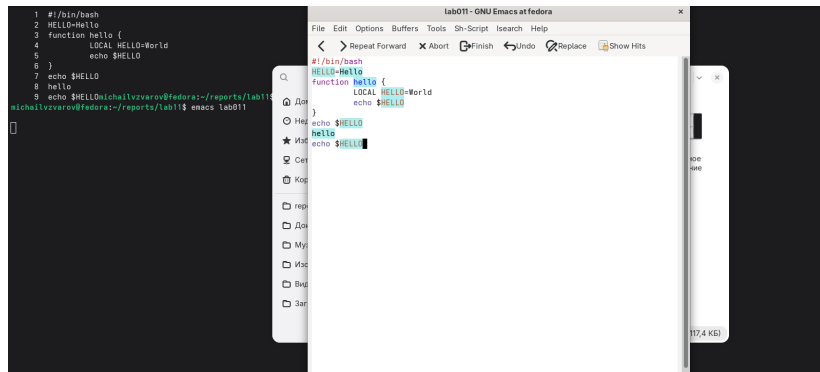
C-s

Other useful commands:

C-r Search backward

M-% Query replace

C-g Cancel the current operation



File Verification

The file contents were checked from the terminal with:

```
cat -n lab011
```

This made it possible to verify the final contents after editing.

```
=== Final file ===
1 #!/bin/bash
2 HELLO=Hello
3 function hello {
4     LOCAL_HELLO=World
5     echo $HELLO
6 }
7 echo $HELLO
8 hello
9 echo $HELLO
michailvzarev@fedora:~/reports/lab11$
```

Figure 5: Final file contents

Final Verification

At the end of the laboratory work, the working directory, available files and final contents were checked.

```
=== Working directory ===  
/home/mikhailvzvarov/reports/lab11  
  
=== Files ===  
wroto 8,0K  
drwxr-xr-x. 1 mikhailvzvarov mikhailvzvarov 734 сен 4 13:34 images  
-rw-r--r--. 1 mikhailvzvarov mikhailvzvarov 107 сен 4 13:33 lab011  
-rw-r--r--. 1 mikhailvzvarov mikhailvzvarov 106 сен 4 12:58 lab011-  
  
=== File contents ===  
1 #!/bin/bash  
2 HELLO=Hello  
3 function hello {  
4     LOCAL_HELLO=World  
5     echo $HELLO  
6 }  
7 echo $HELLO  
8 hello  
9 echo $HELLO  
mikhailvzvarov@fedora:~/reports/lab11$
```

Figure 6: Final verification

Results

During the laboratory work:

- ▶ a file was created and edited in Emacs;
- ▶ keyboard commands for editing and navigation were practiced;
- ▶ active buffers were examined;
- ▶ the frame was split into multiple windows;
- ▶ forward text search was performed;
- ▶ the final file was verified in the terminal.

Conclusion

The basic features of the Emacs text editor were studied.

Practical skills were obtained in:

- ▶ editing text;
- ▶ using keyboard shortcuts;
- ▶ working with buffers;
- ▶ managing multiple windows;
- ▶ searching text.

The objective of the laboratory work was achieved.

Thank You

Questions?